

NICOLÁS SANTIAGO DÍAZ SANTOS

Full Stack Software Engineer | React • Next.js • Node.js • NestJS

Cali, Colombia • [LinkedIn](#) • [GitHub](#) • [Portfolio](#) • diazsantosnicolas10@gmail.com

Software Engineer with full-stack experience across digital products and enterprise applications, specializing in the TypeScript, React, Next.js, and Node.js/NestJS ecosystem. Combining product development, API design, and software architecture with expertise in 3D/WebGL and multimodal AI integration. Focused on building end-to-end maintainable solutions, from user interface and business logic to persistence, asynchronous processing, and cloud deployment.

Core Stack & Practices: React · Next.js · TypeScript · Node.js · NestJS · PostgreSQL · Three.js · AWS · Docker

PROFESSIONAL EXPERIENCE

Volio

Remote

Founder & Lead Full Stack Engineer

May 2026 – Present

Platform: volio-studio.com • Interactive 3D Demo: [View demo](#)

- Designed and implemented the end-to-end architecture of an interactive digital album SaaS platform, covering frontend, backend, persistence, and deployment.
- Developed modular backend services with NestJS and asynchronous task processing using Redis and BullMQ, deployed in Docker containers on AWS EC2.
- Implemented the data layer with PostgreSQL on Neon and multimedia object storage with Cloudflare R2, optimizing secure per-user asset management and distribution.
- Built interactive 3D experiences with Three.js and React Three Fiber, including dynamic geometry video playback and mobile rendering optimization.
- Developed a 2D canvas editor assisted by multimodal AI models, engineering prompts and Structured Outputs to transform photos into fully editable, structured layouts.
- Integrated payments, stateless authentication, and webhooks, designing workflows tailored for asynchronous processing and inter-service communication.

BluePixel Software

Mexico (Remote)

Full Stack Developer Mid

March 2025 – May 2026

- Designed and implemented full-stack features for international enterprise clients including Bimbo, Philip Morris, and Magnicharters, contributing across frontend, backend, and external service integrations.
- Developed the MVP for Bimbo, working directly with stakeholders to transform business requirements into production solutions using React, TypeScript, and Node.js.
- Implemented 20+ endpoints and full-stack features for airline systems, adapting existing services to new business logic and functional requirements.
- Participated in cloud deployment and environment management using Microsoft Azure and Azure DevOps, collaborating on CI/CD workflows, application configuration, and service delivery.
- Built interactive web experiences integrating WebGL and Three.js into corporate applications.

BDE

Mexico (Remote)

Frontend Developer

October 2024 – March 2025

- Developed responsive interfaces and reusable components with React, TypeScript, and Tailwind CSS for administrative and analytics dashboards.
- Implemented interactive data visualizations, converting business metrics into intuitive monitoring and decision-support tools.

Universidad Autónoma de Occidente

Cali, Colombia

Full Stack Developer

January 2024 – June 2024

- Built end-to-end features for an institutional academic management platform using React, Node.js, Sequelize, and MySQL.
- Applied the MPLU+A methodology across requirements gathering, prototyping, and usability validation to optimize administrative workflows.

EDUCATION

UNIVERSIDAD AUTÓNOMA DE OCCIDENTE

Bachelor's Degree in Multimedia Engineering

Honors: Academic Excellence Award Recipient

Cali, Colombia

October 2024

TECHNICAL SKILLS

Frontend: React, Next.js (App Router), TypeScript, JavaScript, Tailwind CSS, HTML5/CSS3

Backend: Node.js, NestJS, REST APIs, Redis, BullMQ

3D & Web Graphics: Three.js, React Three Fiber, WebGL

Databases: PostgreSQL (Neon), MySQL, Prisma, Sequelize

Cloud & DevOps: AWS (EC2), Microsoft Azure, Azure DevOps, Docker, CI/CD

Architecture: Hexagonal Architecture, SOLID, Design Patterns

Applied AI: Multimodal LLM APIs, Structured Outputs, Prompt Engineering

Languages: Spanish (Native), English (B2 – Professional/Technical)